TTP223 / TTP223B Capacitive Touch Sensor Module

Dimensional Guide, Solder Jumper Truth Table, & Enclosure Integration

TOUCH SENSOR | MOMENTARY / TOGGLE | 2.0V – 5.5V

1. OPERATIONAL OVERVIEW & SPECIFICATIONS

The TTP223 module is a solid-state capacitive touch switch designed to replace traditional mechanical pushbuttons. When a human finger approaches or touches the sensing pad, the change in capacitance triggers the onboard Tontek TTP223-BA6 IC, driving the digital output line.

Parameter	Specification
Operating Voltage	2.0V to 5.5V DC (3.3V native for Pico)
Standby Current	< 3.0 μA (Low Power Mode at 3V)
Response Time	60 ms max in low power mode (220 ms max standby)
Output Drive Current	~8 mA sink / ~4 mA source at VDD = 3V
Sensing Medium	Can detect touch through plastic, acrylic, wood, glass (up to 3–5 mm thickness)
Onboard Indicator	Surface mount red LED illuminates upon touch

2. SOLDER JUMPER CONFIGURATION MATRIX (A & B PADS)

On the back of the red PCB are two pairs of solder pads labeled **A** and **B**. Bridging these pads with solder changes the output logic and behavior:

Jumper A	Jumper B	Operational Mode	Output State (Default → Touched)
Open (0)	**Open (0)**	**Momentary (Direct)** *(Default)*	LOW (0V) → **HIGH (3.3V)**
Short (1)	**Open (0)**	**Momentary Inverted**	HIGH (3.3V) → **LOW (0V)**
Open (0)	**Short (1)**	**Toggle (Latching Switch)**	Toggles HIGH / LOW on each touch
Short (1)	**Short (1)**	**Toggle Inverted**	Toggles LOW / HIGH on each touch

3. MECHANICAL DIMENSIONS & PAD GEOMETRY

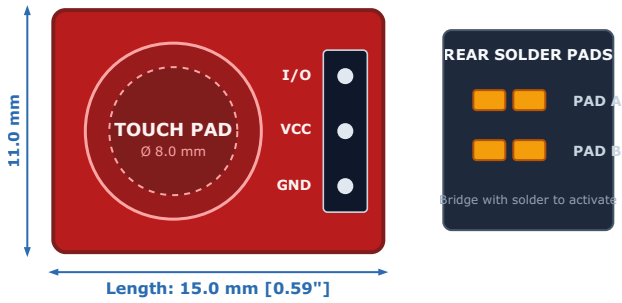


Figure 1: TTP223 Mini Touch Module Geometry, Pinout, and Rear Configuration Jumpers.

4. ENCLOSURE & HIDDEN SENSOR MOUNTING NOTES

Invisible Touch Surfaces: Affix the front touch pad flush against the interior wall of your 3D print or acrylic enclosure using thin double-sided tape. Do not use metal enclosures. Sensitivity remains responsive through acrylic or PLA up to **3.0 mm – 4.0 mm** thickness.

5. PIN WIRING TO RASPBERRY PI PICO

- `VCC` → Pico 3V3 (Pin 36)
- `GND` → Pico GND (Pin 38)
- `I/O` → Any digital GPIO (e.g., GP14 or GP15)

6. CIRCUITPYTHON READING CODE

```
import time, board, digitalio

touch = digitalio.DigitalInOut(board.GP14)
touch.direction = digitalio.Direction.INPUT
# Default mode outputs HIGH when touched:
while True:
    if touch.value:
        print("TOUCH DETECTED!")
    time.sleep(0.05)
```